

THE JOIN OF TWO PROJECTIVE SPACES: ISOTROPY QUOTIENTS AND THE TEST COMPARISON FOR DE MORGAN CUBICAL SETS

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ABSTRACT. A companion paper re-opened the question whether the cubical-type model structure on De Morgan cubical sets presents spaces. We first prove the upper bound: on any totally aspherical test cube category the cubical-type weak equivalences are contained in the test weak equivalences, by a one-paragraph localizer argument; on cartesian cubes the inclusion is strict, by Coquand’s symmetric square. We then test sharpness on the isotropy quotients of the De Morgan square. Quotients by a cyclic group whose generator fixes a cell contract in both structures, by an explicit equivariant interpolation that connections make available; quotients by a freely acting group are $K(H, 1)$ in both structures. For the two remaining groups — the Klein group generated by the transposition and the double negation, and the full hyperoctahedral group — the category of elements of the quotient is computed exactly: for the Klein group it is the *join* $\mathbb{RP}^\infty \star \mathbb{RP}^\infty$, simply connected with third homology $\mathbb{Z}/2$, so the quotient is contractible in neither structure; a machine reachability census over twisted-symmetric cells gives an independent syntactic proof. Consequently every known candidate witness agrees, and no automorphism quotient of the square separates the two localizers. We conjecture that the agreement is general, and isolate the presheaf-level obstacle: the geometric realization of these quotients is contractible, and every stabilized cell is a face of a free cell, so neither cellular computation nor naïve stratification sees the answer.

1. INTRODUCTION

Let $\square_{\text{DM}}$ be the De Morgan cube category — the Lawvere theory of De Morgan algebras, restricted to the free algebras $\text{DM}(n)$ on the interval generators — and let $\widehat{\square_{\text{DM}}}$ be its presheaf category of *De Morgan cubical sets*, the site of the CCHM model of cubical type theory [5]. Two Quillen model structures live on $\widehat{\square_{\text{DM}}}$ with the same cofibrations (all monomorphisms): the *cubical-type* model structure, whose fibrations are the uniform Kan fibrations of cubical type theory, with weak equivalences $\mathcal{W}_{\text{type}}$; and the *test*

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model structure of Cisinski’s theory of test categories [4], whose weak equivalences W_{test} are the maps f such that $\text{el}(f)$ is a weak equivalence of categories, where el denotes the category of elements. Since $\square_{\text{DM}}$ is a strict test category [2, 7], the test structure presents the homotopy theory of spaces. Whether the type structure does — that is, whether $W_{\text{type}} = W_{\text{test}}$ — is the *test comparison problem* for cubical type theory.

For cartesian cubes the answer is negative: Coquand’s symmetric square Q is test-trivial but not type-contractible [6, 1]. For De Morgan cubes the folklore witness was the reversal quotient $L = \mathbb{I}/(x \sim \neg x)$; a companion paper [10] proved L is a $K(\mathbb{Z}/2, 1)$ in the type structure, and its sequel [11] showed the same holds on the test side ($\text{el}(L) \simeq B\mathbb{Z}/2$, by freeness of the reversal action on cells), so the witness is *void* and the De Morgan comparison problem is open.

This paper reports three groups of results.

The upper bound. Theorem 3.1: for every totally aspherical test cube category whose type structure is generated by pushout products of endpoint inclusions with monomorphisms,

$$W_{\text{type}} \subseteq W_{\text{test}}.$$

The proof is a one-paragraph localizer argument. Combined with Coquand’s square, it shows the inclusion is *strict* for cartesian cubes; and its contrapositive — test-nontrivial implies type-nontrivial — becomes the engine of the second half of the paper.

Isotropy quotients. The remaining question is the lower bound, equivalently: is every test-trivial map type-trivial? A separating witness must be test-trivial and type-nontrivial. All known candidates are quotients $\square^n/H$ by subgroups H of the hyperoctahedral automorphism group [11], so we compute both sides for *all* subgroups of $\text{Aut}(\square^2) = G_2$ (dihedral of order 8), and for a representative three-dimensional family:

- *Fixed-stratum quotients* (Theorem 4.1): if a cyclic group’s generator fixes at least one cell, the quotient is contractible in the type structure — an explicit equivariant interpolation, available because connections let one solve the equivariance constraints around the cycle — hence also in the test structure, by Theorem 3.1. This uniformly contracts the symmetric square $\square^2/\langle \text{sw} \rangle$, the twisted square $\square^2/\langle \mathbf{g} \rangle$, and a $\mathbb{Z}/3$ quotient of the cube.
- *Free quotients* (Theorem 5.1): if H acts freely on cells, the quotient is a $K(H, 1)$ in the type structure, matching $\text{el}(\square^n/H) \simeq BH$ [11] on the nose.
- *Mixed quotients* (Theorem 6.1): for the Klein group $K = \{1, \text{sw}, \mathbf{n}, \mathbf{g}\}$ (transposition, double negation, and their product) no cell is fixed by all of K , and the strict quotient $\text{el}(\square^2/K)$ is computed exactly:

it is the join

$$\mathrm{el}(\square^2/K) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2 = \mathbb{RP}^\infty \star \mathbb{RP}^\infty,$$

simply connected with $H_3 = \mathbb{Z}/2$: contractible in *neither* structure. The same method computes the full hyperoctahedral quotient. A twisted reachability census — 72 symmetric squares, of which 66 do contract and the generic one is isolated — gives an independent, purely syntactic proof of type-noncontractibility in the style of [6, 10].

Agreement. Every candidate tested — the reversal circle, the symmetric and twisted squares, all free quotients, the Klein and hyperoctahedral quotients, the $\mathbb{Z}/3$ cube — has the *same* answer in both structures (Table 1). In particular no automorphism quotient of the square separates the localizers (Corollary 8.1). We conjecture the agreement is general (Conjecture 8.2), identify why the mixed-case type computation is genuinely hard (the realization is contractible and every stabilized cell is a face of a free cell, Remark 8.3), and record along the way that the De Morgan cube category is *not* an Eilenberg–Zilber category (Proposition 8.4) — the Kleene midpoint has no (split epi, mono) factorization — which is why elegance-style techniques are unavailable on this site.

All machine verifications are reproduced by `scripts/isotropy.py` in the artifact [12].

2. TWO MODEL STRUCTURES

We write $\mathrm{DM}(n)$ for the free De Morgan algebra on $x_1, \dots, x_n$; $\square_{\mathrm{DM}}$ has objects $[n]$ and $\square_{\mathrm{DM}}([m], [n]) = \mathrm{DM}(m)^n$. Recall $|\mathrm{DM}(1)| = 6$, $|\mathrm{DM}(2)| = 168$, $|\mathrm{DM}(3)| = 7,828,354$; elements are monotone functions on the $2n$ -coordinate *literal cube*, with $\neg$ acting by complement-and-swap within literal pairs [9].

The *cubical-type model structure* on $\widehat{\square_{\mathrm{DM}}}$ has cofibrations the monomorphisms and fibrations the maps with the right lifting property against all pushout products $\delta_e \hat{\otimes} m$ of an endpoint inclusion $\delta_e : 1 \rightarrow \mathbb{I}$ ($e \in \{0, 1\}$) with a monomorphism m ; this is the Cisinski–Sattler construction underlying the CCHM model [5, 8]. Its weak equivalences are $\mathbf{W}_{\mathrm{type}}$. The *test model structure* has the same cofibrations and weak equivalences $\mathbf{W}_{\mathrm{test}} = \{f : \mathrm{Nel}(f) \text{ a simplicial weak equivalence}\}$; it exists and presents spaces because $\square_{\mathrm{DM}}$ is a strict — indeed totally aspherical — test category [4, 7, 2].

An *A-localizer* [4, §1.4] is a class $\mathbf{W}$ of maps of $\widehat{A}$ containing the trivial fibrations, satisfying two-out-of-three, and whose monomorphic members are closed under pushout, transfinite composition, and retract. $\mathbf{W}_{\mathrm{test}}$ is an *A-localizer*, and $\mathbf{W}_{\mathrm{type}}$ is one as well [11]; $\mathbf{W}_{\mathrm{min}}$ denotes the minimal localizer.

$\mathrm{Aut}_{\square_{\mathrm{DM}}}([n]) = G_n$, the hyperoctahedral group $\Sigma_n \ltimes (\mathbb{Z}/2)^n$, acting on cells $(a_1, \dots, a_n) \in \mathrm{DM}(m)^n$ by permuting coordinates and applying $\neg$ [11]. For $n = 2$ we write $\mathrm{sw}(a, b) = (b, a)$, $\mathrm{n}(a, b) = (\neg a, \neg b)$, $\mathrm{g} = \mathrm{swn} : (a, b) \mapsto$

$(\neg b, \neg a)$, and $n_1(a, b) = (\neg a, b)$; $G_2 = \langle \text{sw}, n_1 \rangle$ is dihedral of order 8 and $K = \{1, \text{sw}, n, g\}$ is a Klein four-group.

3. THE UPPER BOUND

Theorem 3.1. *Let A be a totally aspherical test cube category (De Morgan, Kleene, Boolean, or cartesian) whose cubical-type model structure is cofibrantly generated with generating trivial cofibrations $\{\delta_e \otimes m : m \text{ mono}, e \in \{0, 1\}\}$. Then $\mathcal{W}_{\text{type}} \subseteq \mathcal{W}_{\text{test}}$.*

Proof. $\mathcal{W}_{\text{test}}$ is an A -localizer. (i) Since A is totally aspherical, el preserves finite products up to weak equivalence [4, §1.6], and $\text{el}(\mathbb{I})$ has a terminal object; hence the projection $X \times \mathbb{I} \rightarrow X$ is in $\mathcal{W}_{\text{test}}$ for every X , and by two-out-of-three so is each end inclusion $X \times \{e\} \hookrightarrow X \times \mathbb{I}$ — a monomorphism. (ii) For a monomorphism $m : X \rightarrowtail Y$, the map $\delta_e \hat{\otimes} m$ is the comparison $Y \cup_X (X \times \mathbb{I}) \rightarrow Y \times \mathbb{I}$. The inclusion $Y \hookrightarrow Y \cup_X (X \times \mathbb{I})$ is a pushout of the monomorphic test equivalence $X \hookrightarrow X \times \mathbb{I}$, hence in $\mathcal{W}_{\text{test}}$ by the localizer axioms; the end inclusion $Y \hookrightarrow Y \times \mathbb{I}$ is in $\mathcal{W}_{\text{test}}$ by (i); so $\delta_e \hat{\otimes} m \in \mathcal{W}_{\text{test}}$ by two-out-of-three. (iii) The trivial cofibrations of the type structure are retracts of transfinite compositions of pushouts of the generators, all monomorphisms, so they lie in $\mathcal{W}_{\text{test}}$; the trivial fibrations lie in every localizer; and an arbitrary map of $\mathcal{W}_{\text{type}}$ factors as a trivial cofibration followed by a trivial fibration. $\square$

Corollary 3.2. *On cartesian cubical sets $\mathcal{W}_{\text{type}} \subsetneq \mathcal{W}_{\text{test}}$: the inclusion holds by Theorem 3.1, and it is strict because Coquand’s symmetric square is test-trivial [1] but not type-trivial [6].*

Remark 3.3. The contrapositive of Theorem 3.1 reads: a map that is *not* a test equivalence is *not* a type equivalence. All type-noncontractibility results below can be read off from element-category computations this way; we also give an independent syntactic proof (§7) in the tradition of [6, 10], which is the only technique that works *inside* the type structure.

4. QUOTIENTS THAT CONTRACT: THE FIXED-STRATUM RETRACTION

Theorem 4.1. *Let $h \in G_n$ generate a cyclic group $C = \langle h \rangle$ and suppose h fixes at least one cell of $\square^n$. Then $\square^n/C \rightarrow 1$ is in $\mathcal{W}_{\text{type}}$ (hence in $\mathcal{W}_{\text{test}}$).*

Proof. We give the argument for the order-3 element $h(a, b, c) = (\neg c, \neg a, b)$ of G_3 , whose fixed cells are the twisted diagonal $(a, \neg a, \neg a)$; the general case is identical, solving around each cycle of h (fixed cells exist precisely when every cycle has even negation weight, and the cycle-symmetrized meets provide invariant terms).

Write σ for the substitution $(x, y, z) \mapsto (\neg z, \neg x, y)$, so that precomposition of the generic cell with σ equals the h -translate. A homotopy $G = (G_1, G_2, G_3) : \square^3 \times \mathbb{I} \rightarrow \square^3$ descends to $\square^3/C \times \mathbb{I} \rightarrow \square^3/C$ iff $G \circ \sigma = h \circ G$, that is,

$$G_1 \circ \sigma = \neg G_3, \quad G_2 \circ \sigma = \neg G_1, \quad G_3 \circ \sigma = G_2.$$

Set $G_3 := \neg(G_1 \circ \sigma)$ and $G_2 := \neg(G_1 \circ \sigma^2)$: the first two equations hold by construction and the third follows from $\sigma^3 = \text{id}$. *The component G_1 is therefore free*, subject only to its face conditions — and these propagate automatically: if $G_1|_{t=0} = x$ then $G_2|_{t=0} = \neg(x \circ \sigma^2) = \neg\neg y = y$ and $G_3|_{t=0} = z$; if $G_1|_{t=1} = m$ with $m \circ \sigma = m$, then $G|_{t=1} = (m, \neg m, \neg m)$, a fixed cell. Take the invariant term $m = x \wedge \neg y \wedge \neg z$ and the standard interpolation

$$G_1 = (x \wedge \neg t) \vee (m \wedge t) \vee (x \wedge m).$$

All identities — $\sigma^3 = \text{id}$, invariance of m , the three equivariance equations, the six face conditions, and monotonicity — are verified on all 256 literal points [12]. The homotopy contracts $\square^3/C$ onto its fixed stratum, a copy of $\mathbb{I}$ (the twisted diagonal embeds: its cells are fixed, so the quotient identifies nothing there), which contracts to a vertex by $a \mapsto a \vee s$ within the stratum. $\square$

Corollary 4.2. $\square^2/\langle \text{sw} \rangle$ and $\square^2/\langle \mathbf{g} \rangle$ are contractible in both structures: for sw take the invariant term $m = x \wedge y$ (recovering the connection contraction $q(s, t) \mapsto q(s \vee z, t \vee z)$), for $\mathbf{g}$ take $m = x \wedge \neg y$.

Remark 4.3. This is exactly what fails for cartesian cubes: Coquand’s square is $\square^2/\langle \text{sw} \rangle$ over a site with no connections, so no interpolation G_1 exists and the quotient is a genuine type-theoretic obstruction [6]. Over De Morgan cubes the connections dissolve every *cyclic* fixed-stratum obstruction; what survives is the mixed case of §6, where no cell is fixed by the whole group and no retraction target exists.

5. FREE QUOTIENTS

Call $H \leq G_n$ *cell-free* if no nontrivial element fixes a cell of $\square^n$ at any level; by the freeness theorem of [10, 11] this holds for every subgroup generated by elements with a cycle of odd negation weight (for $n = 2$: $\langle \mathbf{n} \rangle$, $\langle \mathbf{n}_1 \rangle$, $\langle \mathbf{n}_1 \text{sw} \rangle \cong \mathbb{Z}/4$, $\langle \mathbf{n}_1, \mathbf{n}_2 \rangle$, and conjugates).

Theorem 5.1. *If H is cell-free then $\square^n/H$ is a $K(H, 1)$ in the cubical-type model structure. Combined with $\text{el}(\square^n/H) \simeq BH$ [11, Lem. elBH], the two structures agree on the nose on all cell-free quotients.*

Proof sketch. The syntactic fibrant replacement $T(-)$ of [10, §4] commutes with cell-free quotients: $T(Y)/H \cong T(Y/H)$, because a nontrivial h fixing a comp-cell would fix its base data, and the base cells are free — the induction of [10, Prop. 4.3] generalizes verbatim from $\mathbb{Z}/2$ to arbitrary H . Hence the fibrant replacement of $\square^n/H$ is R/H with $R = T(\square^n)$ contractible and fibrant. The projection $R \rightarrow R/H$ has unique lifting of open boxes (boxes are connected and the action is free), so R/H is fibrant and the projection is a principal H -covering fibration with contractible total space: transport on the fiber is free and transitive, giving $\pi_1 = H$, and the higher homotopy of R/H vanishes with that of R . $\square$

6. THE MIXED QUOTIENTS: THE JOIN

Two subgroups of G_2 remain: the Klein group K and G_2 itself. No cell of $\square^2$ is fixed by all of K : a fixed cell would satisfy $b = a$ (from sw) and $a = \neg a$ (from n), which is impossible. The stabilizer of a cell (a, b) is $\langle \text{sw} \rangle$ if $b = a$ (the *diagonal stratum*), $\langle \text{g} \rangle$ if $b = \neg a$ (the *antidiagonal stratum*), and trivial otherwise (the *free stratum*).

Theorem 6.1. $\text{el}(\square^2/K) \simeq B\mathbb{Z}/2 \star B\mathbb{Z}/2$, the join of two infinite projective spaces: simply connected, with $\tilde{H}_2 = 0$ and $H_3 = \mathbb{Z}/2$. Moreover $\text{el}(\square^2/G_2) \simeq \text{hocolim}(B\mathbb{Z}/2 \leftarrow BK \rightarrow BG_2)$, which has fundamental group $\mathbb{Z}/2$.

Proof. Write $C = \text{el}(\square^2)$, the slice $\square_{\text{DM}}/[2]$, a contractible category (terminal object $T = \text{id}$) on which K acts by postcomposition; the category of elements of the quotient presheaf is the strict quotient: $\text{el}(\square^2/K) \cong C/K$.

Step 1 (sieve structure). Along any morphism $[f] \rightarrow [g]$ of C/K one has $\text{Stab}(g) \subseteq \text{Stab}(f)$: if $g \circ u = f$ and $k \in \text{Stab}(g)$ then $kf = kg u = gu = f$ (the group is abelian, so stabilizers are well-defined on orbits). There are no morphisms from the free stratum into a stabilized one, and none between the two stabilized strata: a morphism into the diagonal stratum forces equal components on a K -orbit that has none (verified also by machine over all 23,040 small-level composites, zero violations [12]). So the two stabilized strata form a sieve $S = D \sqcup A$ and the free stratum a cosieve U , and C/K is the collage of the hom profunctor $W : S^{\text{op}} \times U \rightarrow \text{Set}$.

Step 2 (the three pieces). The nerve of a collage is the double mapping cylinder $\text{hocolim}(NS \leftarrow N\widehat{W} \rightarrow NU)$, where $\widehat{W}$ is the two-sided category of elements of W (objects: an arrow from a stabilized object to a free one).

- U : the free stratum contains T and is closed under maps out, so it is contractible before quotienting; the K -action on it is free, so $U \simeq BK$.
- D : the diagonal stratum of C is $\text{el}(\mathbb{I}) \simeq *$ via $a \mapsto (a, a)$; sw acts trivially on it and n freely, so $D \simeq B(K/\langle \text{sw} \rangle) = B\mathbb{Z}/2$ — a copy of $\text{el}(L)$ [11]. Likewise $A \simeq B(K/\langle \text{g} \rangle)$.
- $\widehat{W}_D$: it is isomorphic to $\text{Arr}(D_C, U_C)/K$, the quotient of the category of arrows from diagonal to free objects of C . The projection $\text{Arr} \rightarrow D_C$ is a Grothendieck fibration whose fiber over d is the comma $(d \downarrow U_C)$, which has terminal object the unique map $d \rightarrow T$; so Arr is contractible by Quillen's Theorem A, the K -action on it is free (its free-stratum component is), and $\widehat{W}_D \simeq BK$.

Step 3 (the maps, and the join). The functor $\widehat{W}_D \rightarrow U$ is the free quotient of an equivariant map of contractible categories, hence an equivalence $BK \rightarrow BK$; the functor $\widehat{W}_D \rightarrow D$ is equivariant along $K \twoheadrightarrow K/\langle \text{sw} \rangle$, hence the classifying map of that projection. Gluing the two cylinders along the

equivalences leaves

$$\mathrm{el}(\square^2/K) \simeq \mathrm{hocolim}(B(K/\langle \mathrm{sw} \rangle) \longleftarrow BK \longrightarrow B(K/\langle \mathrm{g} \rangle)).$$

Since $K = \langle \mathrm{sw} \rangle \times \langle \mathrm{g} \rangle$, the two legs are the projections $B\mathbb{Z}/2 \times B\mathbb{Z}/2 \rightarrow B\mathbb{Z}/2$, and the homotopy pushout of the two projections of a product is by definition the join. Joins of path-connected spaces are simply connected, and $\tilde{H}_n(X \star Y) \cong \tilde{H}_{n-1}(X \wedge Y)$ gives $\tilde{H}_2 = 0$ and $H_3 = H_1 \otimes H_1 = \mathbb{Z}/2$.

For G_2 the reflection n_1 interchanges the two stabilized strata, which merge into one with $S \simeq B(\langle \mathrm{sw}, n \rangle / \langle \mathrm{sw} \rangle)$, and the interface becomes BK mapping to $U \simeq BG_2$ along the inclusion $K \hookrightarrow G_2$; van Kampen gives $\pi_1 = G_2 / \langle \langle \mathrm{sw} \rangle \rangle = G_2 / K = \mathbb{Z}/2$ (the normal closure of sw contains its n_1 -conjugate g). $\square$

Corollary 6.2. $\square^2/K \rightarrow 1 \notin W_{\mathrm{type}}$ and $\square^2/G_2 \rightarrow 1 \notin W_{\mathrm{type}}$: both quotients are contractible in neither structure.

Proof. Immediate from Theorem 6.1 and the contrapositive of Theorem 3.1. $\square$

7. THE SYNTACTIC CENSUS

Corollary 6.2 passes through the test structure. We record an independent proof of $\square^2/K \rightarrow 1 \notin W_{\mathrm{type}}$ that never leaves the type structure, by the factorization-invariant method of [6, 10]; it is the only known technique of this kind, and it doubles as a verification of the census machinery.

Call a square c of a De Morgan cubical set K -*symmetric* if $c \circ \sigma_{\mathrm{sw}} = c$ and $c \circ \sigma_n = c$, where $\sigma_{\mathrm{sw}}, \sigma_n$ are the corresponding automorphisms of [2] acting by precomposition (for cells of a quotient this is orbit equality, so the condition unfolds to a twisted equation with a twist in K); similarly for cubes, with the extra direction untouched. Let $P''(X)$ be the property: every K -symmetric square of X is connected to a constant square by a finite chain of K -symmetric cubes.

Proposition 7.1. $P''(\square^2/K)$ fails: the K -symmetric squares number 72 and the chain graph decomposes into the component of the generic square (size 3), the component of the constants (size 66), and one further component (size 3); the generic square reaches no constant.

The census enumerates, for each assignment of a twist in K to each of the two generators, the compatible cubes: a twist with a transposition component determines one component of the cube from the other over all of $\mathrm{DM}(3)$, and the purely diagonal twists reduce to face-profile combinations; positive and negative controls (the cyclic subgroups, which must connect by Corollary 4.2, and the cell-free subgroups, which must not, by Theorem 5.1) all behave as theory predicts [12]. That 66 of 72 symmetric squares *do* contract shows the invariant is nondegenerate: the isolation of the generic square is not a formal artifact.

quotient	type structure	test structure
$L = \mathbb{I}/\langle - \rangle$	$K(\mathbb{Z}/2, 1)$ [10]	$B\mathbb{Z}/2$ [11]
$\square^2/\langle \text{sw} \rangle, \square^2/\langle \text{g} \rangle$	$*$ (Cor. 4.2)	$*$ (Thm. 3.1)
$\square^3/\mathbb{Z}/3$ twisted	$*$ (Thm. 4.1)	$*$ (Thm. 3.1)
cell-free H	$K(H, 1)$ (Thm. 5.1)	BH [11]
$\square^2/K$	$\neq *$ (Cor. 6.2, Prop. 7.2)	$\mathbb{RP}^\infty \star \mathbb{RP}^\infty$ (Thm. 6.1)
$\square^2/G_2$	$\neq *$ (Cor. 6.2)	$\pi_1 = \mathbb{Z}/2$ (Thm. 6.1)

TABLE 1. Both structures agree on every tested quotient.

Proposition 7.2. *P'' propagates through the syntactic fibrant replacement: for the factorization $1 \rightarrow T \twoheadrightarrow \square^2/K$ of [10, §4], $P''(1)$ implies $P''(T)$; and if $u : X \rightarrow Y$ has a section, $P''(X)$ implies $P''(Y)$. Consequently $\square^2/K \rightarrow 1 \notin \mathbf{W}_{\text{type}}$.*

Proof sketch. As in [10, Prop. 4.3], with one new ingredient: the substitutions $\sigma_{\text{sw}}, \sigma_n$ are *automorphisms* of the free De Morgan algebra on two generators, so no restriction clause of a comp-cell can fire under them, symmetry of a composition cell is an equality of comp-constructors, and all its data (directions, constraint, tube, base) is itself symmetric. The fresh-direction cube $c(x, y, t) = (v(x, y, t), \text{comp}^{k \rightarrow t}(a, \psi, u, \langle z \rangle v(x, y, z)))$ is then K -symmetric, agrees with the given square at $t = l$, and reduces the inductive height at $t = k$; the induction terminates at cells of the point. Given $\square^2/K \rightarrow 1 \in \mathbf{W}_{\text{type}}$, two-out-of-three makes the vertex $1 \rightarrow \square^2/K$ a weak equivalence, the fibration $T \rightarrow \square^2/K$ is then trivial and has a section, and $P''(1) \Rightarrow P''(T) \Rightarrow P''(\square^2/K)$ contradicts Proposition 7.1. $\square$

8. AGREEMENT, NON-SEPARATION, AND THE CONJECTURE

Corollary 8.1. *No quotient $\square^2/H$, $H \leq G_2$, is a separating witness for $\mathbf{W}_{\text{type}} \subseteq \mathbf{W}_{\text{test}}$ on De Morgan cubical sets: the subgroups of G_2 are exhausted by Corollary 4.2 (cyclic with fixed cells), Theorem 5.1 (cell-free), and Corollary 6.2 (mixed), and in every case the quotient is test-trivial if and only if it is type-trivial.*

Conjecture 8.2. *The homotopy types agree on all isotropy quotients in all dimensions: for every $H \leq G_n$, the type structure assigns to $\square^n/H$ the homotopy colimit of its stabilizer strata — the same diagram that computes $\text{el}(\square^n/H)$. In particular $\mathbf{W}_{\text{type}} = \mathbf{W}_{\text{test}}$ on De Morgan cubical sets.*

Remark 8.3. Two remarks on why the remaining inclusion resists. First, the geometric realization of every quotient above is contractible (a folded square is a triangle), so no cellular or realization argument can see the answer; the phenomenon lives strictly in the difference between strict and homotopy quotients, on both sides. Second, the element-category computation of Theorem 6.1 rests on a stratification that has *no presheaf-level counterpart*: the subpresheaf of $\square^2$ generated by the free cells is all of $\square^2$,

because every diagonal cell (a, a) is a face of the free cell $(a, a \vee (t \wedge \neg t))$ — a Kleene-midpoint phenomenon with no Boolean or cartesian analogue. A type-side computation of the mixed quotients therefore needs a genuinely new device (a presheaf-level collage), which we leave open.

Proposition 8.4. $\square_{\text{DM}}$ and $\square_{\text{KL}}$ are not Eilenberg–Zilber categories: the Kleene midpoint $x \wedge \neg x : [1] \rightarrow [1]$ factors through no (split epimorphism, monomorphism) pair.

Proof. The two points of $\mathbb{I}$ have the same image under $x \wedge \neg x$ (both go to $0 \wedge 1 = 0$), so the map is not a monomorphism; it is not constant either, so a factorization would pass through $[1]$ with a split epimorphism $[1] \rightarrow [1]$ whose image is the midpoint — but a split epimorphism is surjective on the interval algebra, and $x \wedge \neg x$ generates a proper subalgebra. $\square$

This explains from first principles why the elegance-based techniques of [3] require the *relative* theory over these sites, and complements the no-go of [11]: on De Morgan cubes even the absolute Eilenberg–Zilber property is unavailable.

Outlook. Three directions. (1) *The type side of the join.* The natural route to Conjecture 8.2 for the mixed quotients is a presheaf-level double mapping cylinder interpolating between $\square^2/K$ and the join, with the two L -copies (Theorem 5.1 and [10]) as the stabilized ends. (2) *Generation of W_{test} .* The equality $W_{\text{type}} = W_{\text{test}}$ would follow from exhibiting a generating set of the localizer W_{test} consisting of maps already in W_{type} ; over a totally aspherical site the candidates are exactly the quotient maps studied here, which is what Conjecture 8.2 would supply. (3) *Beyond quotients.* Corollary 8.1 closes the two-dimensional quotient route to a counterexample; a separating witness, if any, must be either higher-dimensional with non-abelian mixed isotropy, or not an isotropy quotient at all.

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